

Efficient formation of stratospheric aerosol for climate engineering by emission of condensable vapor from aircraft

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[1] Recent analysis suggests that the effectiveness of stratospheric aerosol climate engineering through emission of non-condensable vapors such as SO₂ is limited because the slow conversion to H₂SO₄ tends to produce aerosol particles that are too large; SO₂ injection may be so inefficient that it is difficult to counteract the radiative forcing due to a CO₂ doubling. Here we describe an alternate method in which aerosol is formed rapidly in the plume following injection of H₂SO₄, a condensable vapor, from an aircraft. This method gives better control of particle size and can produce larger radiative forcing with lower sulfur loadings than SO₂ injection. Relative to SO₂ injection, it may reduce some of the adverse effects of geoengineering such as radiative heating of the lower stratosphere. This method does not, however, alter the fact that such a geoengineered radiative forcing can, at best, only partially compensate for the climate changes produced by CO₂. **Citation:** Pierce, J. R., D. K. Weisenstein, P. Heckendorn, T. Peter, and D. W. Keith (2010), Efficient formation of stratospheric aerosol for climate engineering by emission of condensable vapor from aircraft, *Geophys. Res. Lett.*, 37, L18805, doi:10.1029/2010GL043975.

1. Introduction

[2] It may be possible to engineer an increase in the Earth's albedo by increasing the stratospheric aerosol burden. While such "geoengineering" entails novel environmental and security risks and cannot fully compensate for CO₂-driven warming (e.g. ocean acidification and hydrological cycle changes), it may nevertheless provide an important tool for managing climate risk [Shepherd *et al.*, 2009; Blackstock *et al.*, 2009; Keith *et al.*, 2010]. This has inspired recent research to quantify the unintended consequences of geoengineering [Govindasamy and Caldeira, 2000; Govindasamy *et al.*, 2002, 2003; Matthews and Caldeira, 2007; Rasch *et al.*, 2008a, 2008b; Caldeira and Wood, 2008; Robock *et al.*, 2008; Tilmes *et al.*, 2008, 2009; Trenberth and Dai, 2007; Heckendorn *et al.*, 2009]. If the costs and benefits of a geoengineering scheme are to be

seriously considered, the scheme must be able to compensate for large radiative forcings such as a doubling of CO₂.

[3] The radiative forcing caused by the addition of aerosol to the stratosphere depends very strongly on aerosol size for at least two reasons (Figure 1). First, particle settling speed increases rapidly with radius, and faster settling causes shorter stratospheric lifetimes and therefore a smaller time-integrated radiative forcing per mass of aerosol delivered. Second, the amount of solar radiation scattered back to space depends strongly on the size of the scattering aerosol. Scattering intensity per unit aerosol mass decreases rapidly beyond an optimal radius of ~150 nm. In addition, the aerosol absorbs longwave radiation in the lower stratosphere that is roughly proportional to its mass loading. This heating and the availability of reactive surface area leads to changes in stratospheric chemistry [Heckendorn *et al.*, 2009]. Reducing the geoengineered sulfur mass loading by controlling particle size may reduce lower stratospheric heating and thus some of the unintended adverse impacts in the stratosphere.

[4] Volcanoes inject sulfur into the stratosphere almost exclusively as sulfur dioxide (SO₂), which does not itself condense to aerosols. SO₂ is subsequently oxidized to sulfur trioxide (SO₃) within a few weeks [Rasch *et al.*, 2008a]. SO₃ in turn is almost instantaneously (milliseconds) converted to sulfuric acid (H₂SO₄), see auxiliary material Text S1.¹ H₂SO₄ vapor quickly (hours) condenses onto existing aerosols or forms new aerosols by homogeneous nucleation. Most of the recent analyses have assumed that aerosols in stratospheric-aerosol geoengineering would be produced by an analogous process, i.e. the injection of either SO₂ or H₂S gas that is slowly converted to sulfate aerosol. A recent analysis [Heckendorn *et al.*, 2009] suggests that this method may be ineffective because it produces aerosols that are substantially larger than optimal. The added mass accumulates preferentially on larger pre-existing particles either by direct condensation or by homogeneous nucleation followed by coagulation. The net effect is that radiative forcing increases sub-linearly with sulfur emissions, so a large (relative to previous studies) injection rate of 10 Mt-S/yr at the equator and 20 km produces a radiative forcing of only 1.7 W m⁻², less than half of what is needed to offset the radiative forcing of a CO₂ doubling [Heckendorn *et al.*, 2009].

2. Aerosol Formation From H₂SO₄ in an Aircraft Plume

[5] We explore the possibility of forming sulfate aerosols in an aircraft plume by emission of H₂SO₄ vapor rather than

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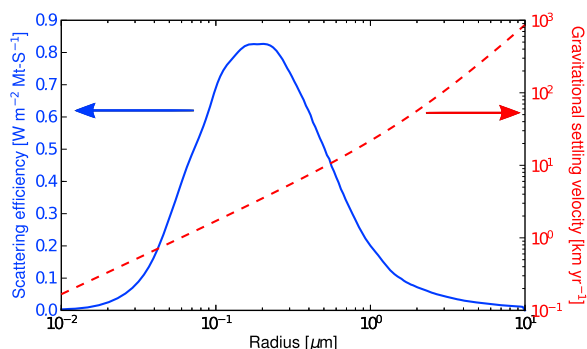


Figure 1. Size dependence of factors that determine the radiative forcing of stratospheric sulphate aerosol. The blue curve and left axis show the solar-band cooling per mass (burden) of sulphate in the stratosphere as a function of aerosol size. The calculation was done assuming monodisperse aerosol and averaging over the solar spectrum. The red curve and right axis show the gravitational settling velocity at a height of 25 km.

SO₂ (Figure S1). (Depending on the injection method, SO₃ vapor may be emitted rather than H₂SO₄, but it is quickly converted to H₂SO₄, see auxiliary material Text S1). Aerosol mass is formed in an aircraft plume when a vapor emitted by the aircraft cools below its condensation point as it mixes with ambient air (similar to the mechanism forming aircraft contrails). More generally, one might exploit the same physics to form a variety of aerosols in the stratosphere by emission of a high-temperature gas-phase substance that has low volatility at ambient temperatures.

[6] To investigate the method, we modeled the formation of sulfate aerosol by the injection of sulfuric acid (H₂SO₄) vapor using an aerosol microphysics model (subsequently called plume model) that follows an expanding parcel in the plume from the time of emission (see auxiliary material Text S1). We integrate the model until the loss of particles by coagulation with ambient particles dominates the self-coagulation, whereupon the AER global 2-D sulfate aerosol model (called AER model and discussed in the next section) becomes the appropriate tool. This hand-off between the models occurs ~2 days after injection (for ambient geoengineered aerosol concentrations of 50 cm⁻³, Figure S3). Although the hand-off time depends on the concentration of particles in the ambient atmosphere, the model results have only a small sensitivity to the exact details of the hand-off time.

[7] Figure 2 shows the sensitivity of the sulfate aerosol size distribution after 2 days in the plume to the rates of H₂SO₄ emission, plume dilution, aerosol nucleation and condensation (see auxiliary material Text S1). The resulting aerosol size distributions are almost entirely controlled by the emission and plume dilution rates (various colored lines in Figure 2). Size distributions are remarkably insensitive to variation in nucleation and condensation rate coefficients (various line types in Figure 2) because all H₂SO₄ vapor has nucleated or condensed within seconds of emission, so that self-coagulation of new particles smoothes any initial differences in the size distribution. These results are consistent with *Turco and Yu* [1999]. Because the results are quite sensitive to the plume dilution rate, uncertainties in the dilution rates tested here will lead to uncertainties in the resultant size distributions. These results demonstrate that

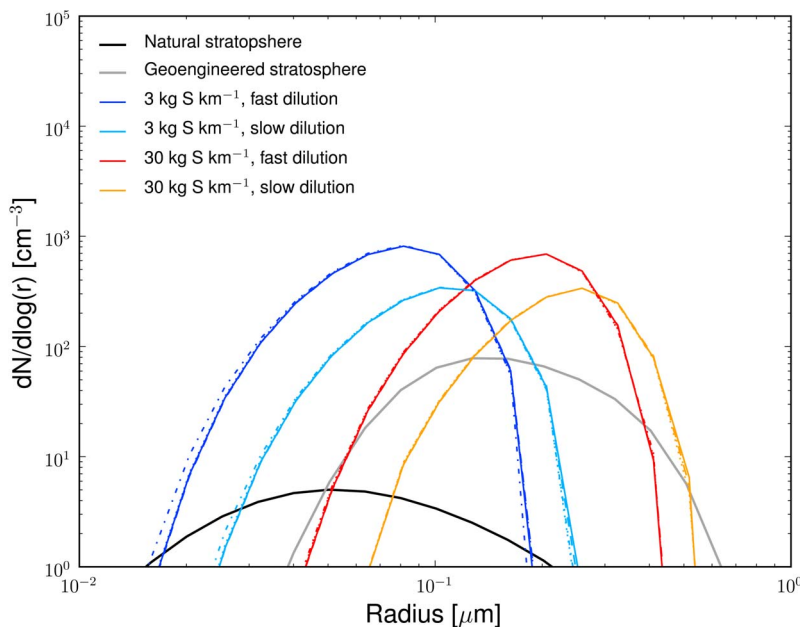


Figure 2. Size distributions of sulfate particles formed in the aircraft plume 2 days after emission of H₂SO₄ derived from the plume model (colored curves) and ambient steady-state size distributions derived from the AER 2D aerosol model at 23 km above the equator without geoengineering (black curve) and with 5 Mt-S yr⁻¹ geoengineering (gray curve, from the 95 nm particle case, described later). For each color, there are four cases plotted for scale factors for nucleation of 1 and 10⁻⁶ and for condensation of 1 and 0.01 (plotted by different line styles, generally overlapping). Ambient particle concentrations with geoengineering of 50 cm⁻³ are used in the plume simulations.

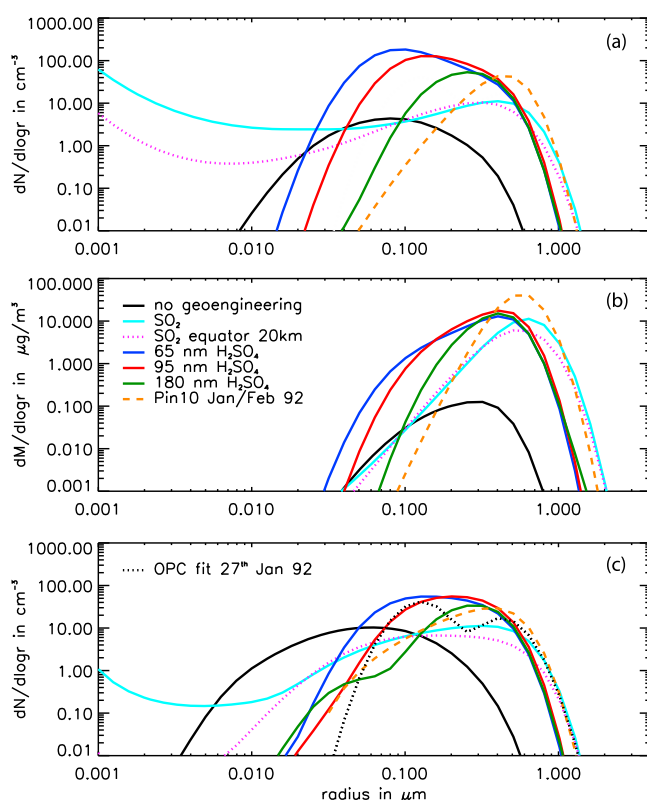


Figure 3. Steady-state annual average aerosol (a) number and (b) mass size distributions at the equator and 23 km and (c) number distributions at 40°N and 17 km predicted by the AER model. Solid black lines: simulations without geoengineering (volcanically quiescent background). Solid colored lines: geoengineering cases with 5 MT S yr⁻¹ emission with emissions spread between 30°S and 30°N and 20 and 25 km. Dashed magenta lines: geoengineering case with 5 MT S yr⁻¹ emission as SO₂ at a single grid point centered at the equator and 20 km [from Heckendorn *et al.*, 2009]. Dashed orange lines: AER model simulation for January–February 1992 following the Mt. Pinatubo eruption. Dashed black line: size distribution fit to measurements by the optical particle counter (OPC) instrument at 41°N in January 1992.

particles of a desired size could be made by controlling the emission rate and possibly the plume dilution rate (if initial dispersion could be controlled through aircraft and injector design).

3. Stratospheric Aerosol and Radiative Forcing

[8] To model the evolution of aerosol in the stratosphere after the condensation plume has formed and relaxed, we use the AER 2-D aerosol model in the same configuration as was used by Heckendorn *et al.* [2009] to model SO₂ injection. Simulations are run for 10 years, at which point steady-state aerosol concentrations have been reached in the stratosphere. We simulate the injection of 2, 5 and 10 megatons of sulfur per year evenly distributed between 30°N and 30°S and 20 and 25 km altitude continuously in time. The size distributions of injected aerosol are obtained from the plume model with 3 kg S km⁻¹ fast and slow dilution cases and the 30 kg S km⁻¹ fast dilution case, resulting in number-median radius of

65, 95 or 180 nm with a lognormal width of 1.5, respectively. We compare with cases in which the same amount of sulfur mass is injected in the same region as SO₂.

[9] Figure 3 shows the predicted steady-state aerosol number and mass size distributions (at the equator and 23 km, and 40°N and 17 km) for 5 Mt-S yr⁻¹ cases. The SO₂-emission cases in Figure 3a show the presence of a nucleation mode that is missing from the H₂SO₄-emission cases. The SO₂-emission cases also have more large ($r > 1 \mu\text{m}$) particles due to condensation of H₂SO₄ onto accumulation-mode particles and coagulation of nucleated particles with the accumulation-mode particles. The peak mass of the particles (Figure 3b) generated by H₂SO₄ injection is at smaller sizes than particles from SO₂ injection. Comparison of these size distributions with Figure 1 shows that the H₂SO₄-generated particles will have longer stratospheric lifetimes and be more effective scatters than the SO₂-generated particles. Figure 3 also shows aerosol size distributions for January–February 1992 following the eruption of Mt. Pinatubo in the Philippines in June 1991. The AER model calculation from Heckendorn *et al.* [2009] is shown at both the equator and 40°N; additionally, size distributions derived from optical particle counter data [Deshler *et al.*, 1993] are shown at 40°N. For the time and grid points shown, the geoengineering calculations with H₂SO₄ emissions yield aerosol particles of smaller size than the volcanic case, while the SO₂-emission cases lead to particles comparable to or larger than the volcanic case.

[10] Figure 4 shows the steady-state stratospheric sulfur burdens and top of atmosphere solar-band flux changes for various injection scenarios. Our SO₂ reference scenario is more efficient than the Heckendorn *et al.* [2009] scenario (“SO₂ equator, 20 km”, replotted here for comparison) because we assume more spatially dispersed SO₂ emissions (30°S–30°N and 20–25 km). Also included for comparison are the stratospheric sulfur burdens computed by Rasch *et al.* [2008b] and the solar flux changes computed by Robock *et al.* [2008], representative of previous modeling that did not include online aerosol microphysics. The sulfur burdens are always higher for the H₂SO₄-injection schemes compared to the SO₂-injection schemes with the same sulfur emission (Figure 4a) due to slower average fall speeds; however, they are still lower than values assumed in previous benchmark studies of geoengineering [Rasch *et al.*, 2008b]. The sulfur burdens from 10 MT yr⁻¹ of H₂SO₄ injection are comparable to the maximum sulfur burden obtained following the Pinatubo eruption, which emitted approximately 10 MT-S (not shown). The SO₂-injection cases are extended to 20 Mt-S yr⁻¹ due to their lower radiative efficiency.

[11] We use a simple top-of-atmosphere solar-band radiative flux calculation (see auxiliary material Text S1) to determine the relative change in the aerosol radiative forcing between the SO₂- and H₂SO₄-injection schemes (Figure 4b) ignoring feedbacks in atmospheric temperature, chemistry, and dynamics (these effects are of secondary importance, see auxiliary material Text S1). The H₂SO₄-injection cases reach a cooling of -4 W m^{-2} (approximately canceling the global-mean warming from a doubling of CO₂) for less than 10 Mt-S yr⁻¹ (approaching previous estimates assumed in the literature [Robock *et al.*, 2008]). On the other hand, the similar SO₂-injection scheme did not reach -4 W m^{-2} until 19 Mt-S yr⁻¹, and the SO₂-injection scheme at the equator

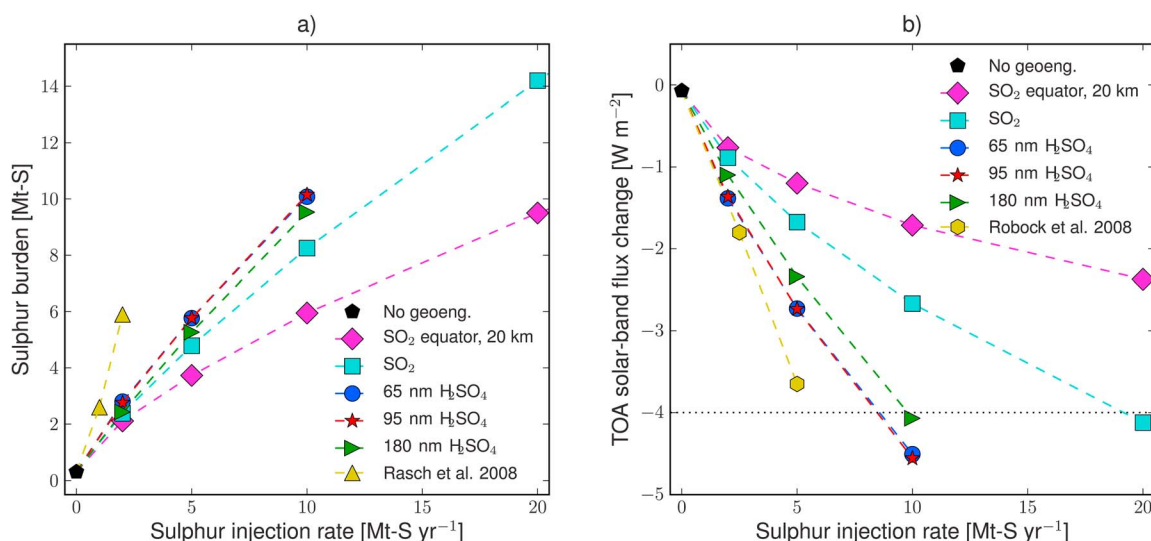


Figure 4. Steady-state (a) stratospheric sulfur burden and (b) top-of-atmospheric solar-band (shortwave) radiative flux change from the stratospheric aerosols as a function of sulfur injection rate. All simulations have emissions evenly distributed between 30°S–30°N and 20–25 km, except results for SO₂ emitted only above the equator (5°S–5°N) at 20 km (19.5–20.5 km). Also included for comparison are the stratospheric sulfur burdens computed by *Rasch et al.* [2008a] (with fixed effective radius of 0.43 μm) and the solar flux changes by *Robock et al.* [2008], both without aerosol microphysics. Black horizontal dotted line in Figure 4b represents the approximate cooling necessary to offset a doubling of CO₂ in the global-mean energy budget.

and 20 km only did not reach this cooling even at 50 Mt-S yr⁻¹ showing the great importance of injection location.

4. Discussion and Conclusions

[12] Our results suggest that emission of H₂SO₄ from aircraft may be used to maintain sulfate aerosol burdens and radiative forcings far more efficiently than can be achieved by continuous SO₂ injection. A 50–60% lower sulfur injection rate was required to cool -4 W m^{-2} for H₂SO₄-injection versus SO₂. (This is a 35% reduction in the total mass brought to the stratosphere if H₂SO₄ is carried rather than SO₂, see injection methods in auxiliary material Text S1.) At -4 W m^{-2} , the stratospheric sulfate mass burden for the H₂SO₄-injection cases is about 40% lower than the SO₂ cases. This will lead to less long-wave warming of the lower stratosphere (proportional to the aerosol mass loading) and thus smaller changes in stratospheric chemistry associated with this warming [Heckendorn et al., 2009]. However, all injection methods yield roughly similar aerosol surface area densities for similar radiative coolings, thus reductions in ozone due to surface-area-catalyzed reactions [Tilmes et al., 2008] do not greatly depend on the injection method explored here and could be substantial in all these cases.

[13] Emission of directly condensable vapors, as proposed here, might also be used to produce non-sulfur aerosols with a variety of compositions without natural analogues [Swihart, 2003]. An advantage of sulfur is that it mimics a well known natural process, but other compounds might give the same radiative forcing with less mass loading and possibly less impact on stratospheric chemistry and dynamics. For example, alumina (Al₂O₃) has about four times as much scattering per unit volume, so it might be possible to use substantially smaller particle fluxes to

achieve the same radiative forcing. Moreover, compared to sulfate, alumina has less infrared absorption and thus would heat the lower stratosphere less, which may produce less ozone loss through this pathway [Pollack et al., 1976; Jackman et al., 1998]. However, chemicals that are less common in nature than sulfate potentially contribute additional risks, both known and unknown.

[14] In order to build understanding of the effectiveness and risks of geoengineering, we first must develop methods that might actually achieve significant negative radiative forcing. Geoengineering by stratospheric aerosol enhancement introduces novel environmental risks and can – at best – partially mask the impacts of greenhouse gases on tropospheric temperatures, whereas issues such as ocean acidification remain unaddressed. This work suggests that the direct formation of aerosols from a plume of low volatility vapors might (a) offer better control of particle size distributions and consequently more efficient negative radiative forcing per unit mass of emission, (b) enable radiative forcings up to -4 W m^{-2} that may be difficult to achieve by continuous emission of SO₂, and finally (c) offer the possibility of using alternative particle compositions that might enable geoengineering with smaller mass loadings and or lower environmental risks [Blackstock et al., 2009]. Beyond the risks arising from the fact that no aerosol geoengineering scheme can fully offset the impacts of greenhouse gases (e.g. ocean acidification and effects on hydrologic cycle), other risks include loss of ozone through heterogeneous chemistry and the impacts of the lower stratospheric warming through absorption of terrestrial radiation [Matthews and Caldeira, 2007; Robock et al., 2008; Tilmes et al., 2008, 2009; Trenberth and Dai, 2007]. Risk assessment needs to balance the negative side effects with the positive effects from the intended cooling, while efforts

towards rapid greenhouse gas emission reductions should not slacken.

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